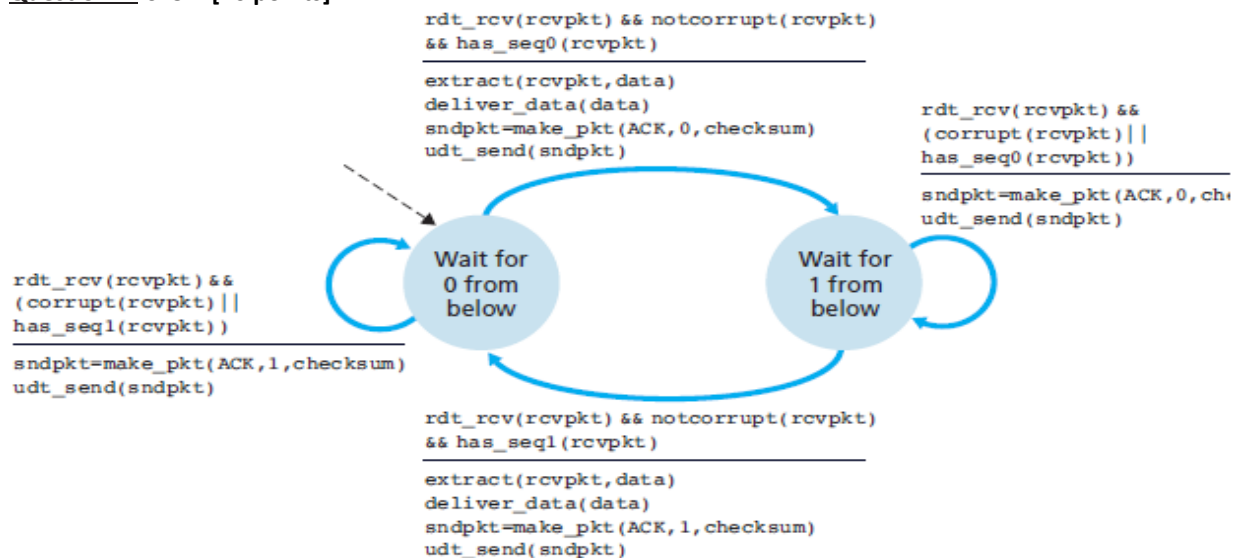
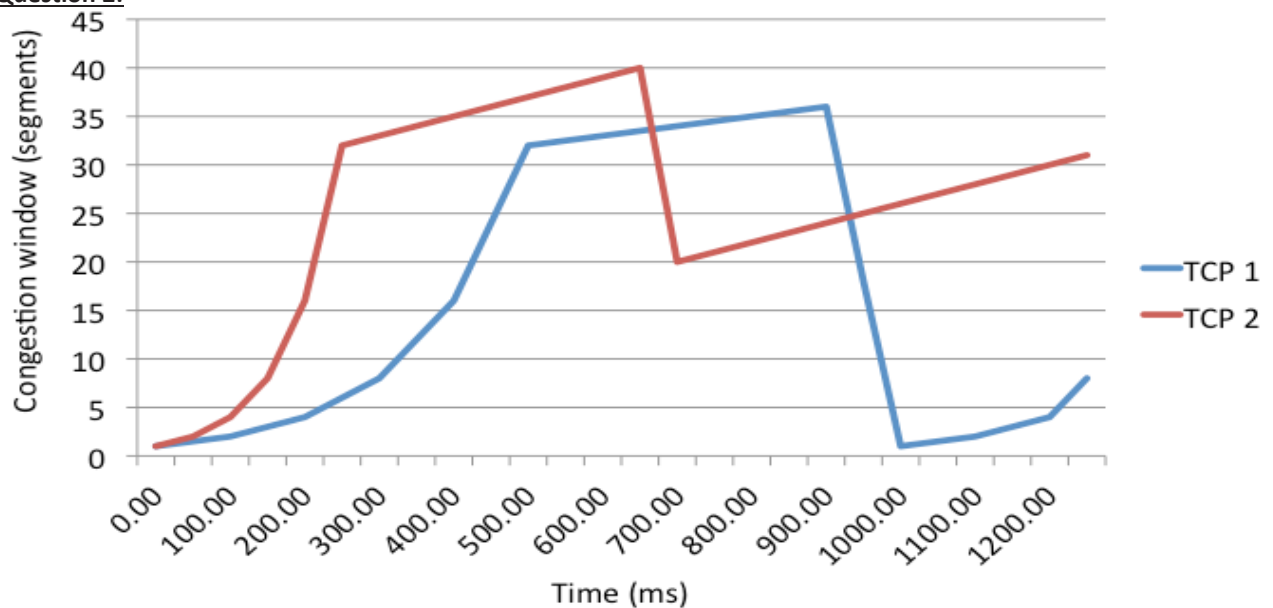


Midterm 2 Examination SOLUTION, Spring 2023
3rd April 2023, 11:30 am – 12:30 pm

Question 1: CLO# [10 points]



Question 2:



A) RTT = 100 ms

$t_0 = 1 \text{ MSS}$
 $t_{100} = 2$
 $t_{200} = 4$
 $t_{300} = 8$
 $t_{400} = 16$
 $t_{500} = 32$
 $t_{600} = 33$
 $t_{700} = 34$
 $t_{800} = 35$
 $t_{900} = 36$

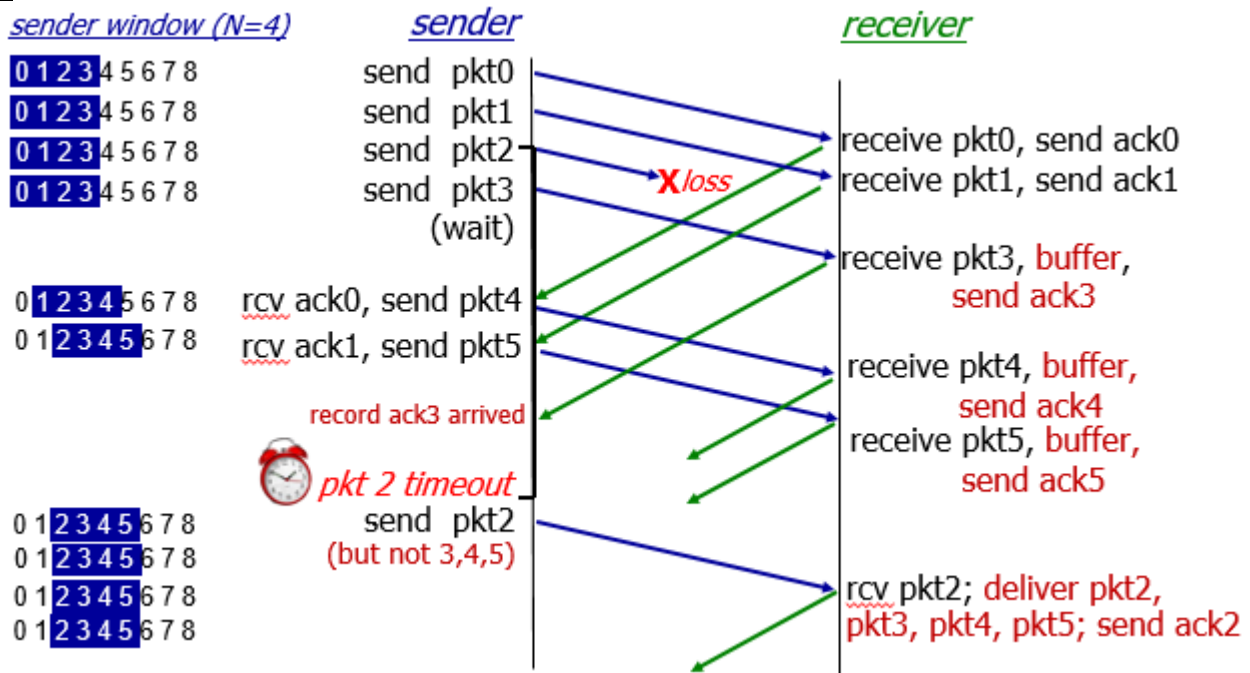
since T1 uses Tahoe triple duplicates will initiate the **cwnd = 1 and ssthresh = 18**

B) RTT = 50 ms

$t_0 = 1 \text{ MSS}$
 $t_{50} = 2$
 $t_{100} = 4$
 $t_{150} = 8$
 $t_{200} = 16$
 $t_{250} = 32$
 $t_{300} = 33$
 $t_{350} = 34$
 $t_{400} = 35$
 $t_{450} = 36$
 $t_{500} = 37$
 $t_{550} = 38$
 $t_{600} = 39$
 $t_{650} = 40$

since T2 uses Reno triple duplicates will initiate fast recovery the **cwnd = 23 and ssthresh = 20**

Question 3:



Question 4:

- TCP slowstart is operating in the intervals [1,6] and [23,26]
- TCP congestion avoidance is operating in the intervals [6,16] and [17,22]
- After the 16th transmission round, packet loss is recognized by a triple duplicate ACK. If there was a timeout, the congestion window size would have dropped to 1.
- After the 22nd transmission round, segment loss is detected due to timeout, and hence the cwnd is set to 1.
- The threshold is initially 32, since it is at this window size that slowstart stops and congestion avoidance begins.
- The threshold is set to half the value of the congestion window when packet loss is detected. When loss is detected during transmission round 16, the congestion windows size is 42. Hence the threshold is 21 during the 18th transmission round.
- The threshold is set to half the value of the congestion window when packet loss is detected. When loss is detected during transmission round 22, the congestion windows size is 29. Hence the threshold is 14.5 ~ 15 during the 24th transmission round.
- During the 1st transmission round, packet 1 is sent; packet 2-3 (2 packets) are sent in the 2nd transmission round; packets 4-7 (4 packets) are sent in the 3rd transmission round; packets 8-15 (8 packets) are sent in the 4th transmission round; packets 15-31 (16 packets) are sent in the 5th transmission round; packets 32-63 (32 packets) are sent in the 6th transmission round; packets 64 – 96 (33 packets) are sent in the 7th transmission round. Thus packet 70 is sent in the 7th transmission round.
- The congestion window and threshold will be set to half the current value of the congestion window (8) when the loss occurred. Thus, the new values of the ssthreshold will be 4, and window will be 7.

Question 5:

Subnet No.	Network Address	Netmask	Host Range	No. Of Hosts
Lahore	12.56.1.0/25	255.255.255.128	12.56.1.1 --- 12.56.1.126	126
Karachi	12.56.1.128/26	255.255.255.192	12.56.1.129 --- 12.56.1.190	62
Islamabad	12.56.1.192/27	255.255.255.224	12.56.1.193 --- 12.56.1.222	30
Peshawar	12.56.1.224/28	255.255.255.240	12.56.1.225 --- 12.56.1.238	14
Faisalabad	12.56.1.240/29	255.255.255.248	12.56.1.249 --- 12.56.1.254	6